

ESSIAW CHARLES JNR

Software Developer & Full-Stack Engineer

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PROFESSIONAL SUMMARY

High-velocity Full-Stack Software Developer specializing in resilient web applications, real-time event systems, and clean UI engineering. Proven track record architecting mission-critical production platforms including cloud restaurant POS/KDS systems, emergency dispatch command software, and secure QR access control. Proficient in TypeScript, Next.js, Supabase, PostgreSQL, Python, and Docker with a disciplined focus on sub-second latency, 100% Core Web Vitals, and modular, testable codebases.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL (PostgreSQL), HTML5, CSS3
Frontend: Next.js (App Router), React, Tailwind CSS, Responsive UI/UX, State Management, Core Web Vitals
Backend & APIs: Node.js, Supabase (Realtime & Auth), RESTful APIs, WebSockets, Python (FastAPI/Flask)
Databases & Cloud: PostgreSQL, Supabase DB, Relational Data Modeling, Docker, Vercel, Render, Git/GitHub
Specializations: Real-Time Event Pipelines, Point of Sale Systems, QR Cryptographic Security, GPS Geolocation, UI/UX

KEY SOFTWARE PROJECTS & PRODUCTION ARCHITECTURE

Sweetbite Food POS & Kitchen Display System (KDS) Active Project

Next.js 14, TypeScript, Supabase Realtime, PostgreSQL, Tailwind CSS

- **Real-Time Pipeline:** Architected a synchronized cloud Point of Sale (POS) and Kitchen Display System (KDS) for food service operations, eliminating manual order slips and kitchen communication delays.
- **Sub-Second Latency:** Leveraged Supabase Realtime WebSocket channels to broadcast order creation, updates, and ticket closure across multiple kitchen stations with sub-second latency.
- **Kitchen UX:** Implemented auditory order alerts, responsive touch-friendly ticket layouts, automated table-routing, and optimistic updates for reliable offline/reconnection handling.

Ghana Emergency Response & Dispatch Telemetry System Live Deployment

TypeScript, JavaScript, Python, PostgreSQL, Geolocation API, WebRTC

- **Rapid Dispatch:** Developed emergency response web platform enabling instant geolocation-tagged SOS distress calls for rapid dispatch and emergency services across Ghana.
- **Live Telemetry:** Engineered live coordinate tracking with interactive mapping and instantaneous voice-note transmission, decreasing dispatch response coordination times significantly.
- **Fault Tolerance:** Integrated high-reliability fallback queues to persist emergency alerts under intermittent cellular connectivity.

Patron Housing Access Control & Visitor Management Production Solution

Python, PostgreSQL, Docker, QR Cryptographic Passports, Modern CSS

- **Access Security:** Engineered automated residential access control platform handling digital visitor issuance, security gate check-in, and cryptographic QR code validation.
- **Infrastructure:** Built administrative dashboards with real-time entry/exit logs, resident authorization workflows, and Dockerized deployment on Render cloud infrastructure.

Enterprise Client Platforms (BizConnect & Kelrose Tours) Client Commercial

Modern JavaScript, Accessible UI/UX, SEO Optimization, REST APIs

- **BizConnect:** Delivered enterprise corporate web portal for BizConnect Technologies with zero bundle bloat and 100% Core Web Vitals performance score.
- **Kelrose Tours:** Designed and shipped Kelrose Tours booking platform with curated dynamic tour itineraries, interactive location galleries, and client lead conversion funnels.

ENGINEERING EXPERIENCE & MILESTONES

Full-Stack Systems Engineer & Independent Developer 2024 - Present

Autonomous Software Development & Client Solutions - Accra, Ghana

- **Systems Delivery:** Architect and maintain end-to-end full-stack software products for hospitality, civic security, and commercial clients.
- **Full Lifecycle:** Design relational database schemas, write optimized SQL queries, implement secure authentication, and configure CI/CD deployment pipelines.
- **Client Excellence:** Achieved a 5.0 Google client rating across commercial engagements through disciplined deadlines, transparent communication, and zero-defect deployments.

Commercial Web Engineer & Client Consultant 2023 - 2024

Freelance & Contract Engagements

- **Client Modernization:** Collaborated with SME business leaders to revamp digital presence, modernize legacy frontends, and streamline high-converting customer booking flows.
- **Performance Optimization:** Engineered responsive, accessible layouts following WCAG 2.1 AA standards and optimized mobile responsiveness, resulting in a 40% increase in conversion rates over 40%.

Autonomous Software Apprenticeship & Foundations2022 - 2023

Intensive Systems Engineering & Deliberate Practice (1,500+ Hours)

- **Core Foundations:** Completed rigorous curriculum in data structures, algorithms, asynchronous programming, modern web standards, and cloud containerization.
- **AI-Augmented Engineering:** Pioneered AI-augmented development workflows, pairing with modern developer tools and agents to accelerate development and stress-test code at accelerated speeds.

EDUCATION, CERTIFICATIONS & CONTINUOUS LEARNING

Software Engineering & Full-Stack Web DevelopmentAccra, Ghana

Comprehensive Autonomous Curricula, Open-Source Contributions & Professional Practice

- Continuous mastery in Next.js, Modern TypeScript, PostgreSQL Schema Architecture, and Cloud Systems.